

Figure 9.1 First Generation Computer

9.2. GENERATION OF COMPUTER

First Generation computer (1940-56)

Actually the construction of computer was started well before 19th century. In 1940s the development got momentum in University of Philadelphia, Pennsylvania state, U.S.A, where the world first computer called ENIAC (Electronic Numerical Integral and Computer) was developed and is shown in Figure 9.1.

This computer was developed using the same device which were used to develop a Radio receiver i.e., Valves (vacuum tubes) as shown in Figure 9.2. 18000 valves were used to construct this first computer. Obviously, the size became very big. Its length was 30' and width was 20' (30' x 20'). It dissipated much heat while on working. In order to reduce the heat, many ACs were used. Therefore, the cost of the computer became very high.

In 1950's the semiconductor elements such as silicon and germanium brought into use which replaced the valves. By using these semiconductor devices, the second generation computer was developed.

Second Generation Computer (1956 – 1963)

This was developed by using transistor (semiconductor device). These transistors performed the functions of valves. Hence,



Figure 9.2 Vacuum tube

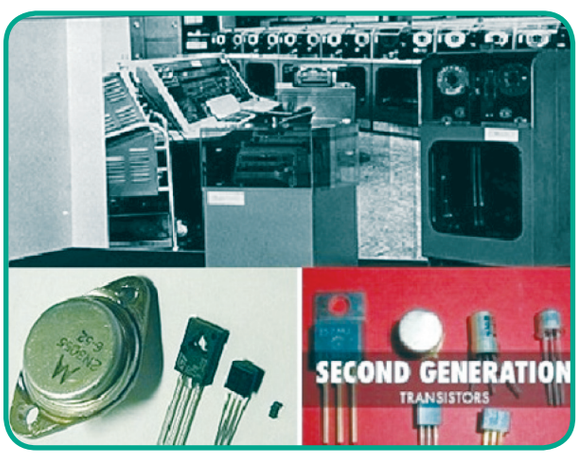


Figure 9.3 Analytical Engine

ELEC_EQUIP_Chapter09.indd 184 11/10/2020 11:53:05 AM